

Unsupervised Spacecraft Anomaly Detection Using a Temporal Autoencoder

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Abstract—Spacecraft telemetry monitoring requires robust, adaptable anomaly detection systems, as hardcoded safety limits frequently fail in dynamic space environments. In this paper, we propose an unsupervised deep learning approach evaluated on the NASA Telemnom dataset, comprising telemetry from the Soil Moisture Active Passive (SMAP) satellite and Mars Science Laboratory (MSL) rover. We implement a 5-block Time-Series Autoencoder combined with Non-parametric Dynamic Thresholding (NDT). Our ablation study demonstrates that combining 1D-CNNs, Long Short-Term Memory (LSTM) networks, Bahdanau-style temporal attention, and skip connections significantly improves the F_1 -score for identifying spacecraft anomalies by capturing both local temporal variances and long-term sequential dependencies.

Index Terms—Anomaly detection, autoencoder, spacecraft telemetry, deep learning, time-series analysis, attention mechanism

I. INTRODUCTION

Modern spacecraft missions generate massive amounts of high-dimensional, non-stationary telemetry data. Monitoring the health of these complex systems traditionally relies on static Out-Of-Limit (OOL) thresholds. However, OOL methods are highly prone to generating false alarms during safe but highly dynamic operations, such as orbital maneuvers or uncommanded safe-mode transitions.

To address the limitations of static bounds, this project implements an unsupervised machine learning solution capable of modeling the nominal operating distribution of spacecraft components. By training a neural network exclusively on normal telemetry, the system forces a reconstruction error when encountering novel anomalous patterns. A dynamic, non-parametric statistical threshold is subsequently applied to flag critical events, adapting to local noise floors and preventing false positives without requiring human intervention.

II. METHODOLOGY

A. Data Preprocessing and Pipeline

The model utilizes the Telemnom dataset, encompassing 55 unique telemetry channels. To maintain temporal integrity and prevent future-data leakage, the time-series data was structured using a chronological sliding window approach with a sequence length of $T = 50$ and a stride of $S = 10$.

The dataset was strictly partitioned chronologically. The final 15% of the normal telemetry sequence was isolated for validation. Test sequences, which contained the labeled

anomalies, were strictly held out to evaluate generalizability. Missing values were imputed using column-wise means, and all features were normalized using a standard scaler fitted exclusively to the training distribution.

B. System Architecture

The anomaly detection model is a multivariate Time-Series Autoencoder consisting of five modular components:

- 1) **1D-CNN**: A one-dimensional convolutional layer (kernel size = 3, padding = 1) coupled with a ReLU activation function to extract initial, localized temporal features.
- 2) **LSTM Encoder**: An LSTM network with a hidden size of 64 to capture long-term sequential dependencies across the temporal dimension.
- 3) **Temporal Attention**: A Bahdanau-style attention mechanism that dynamically weights the importance of specific time-steps. The context vector c_t is computed as a weighted sum of the encoder hidden states h_i :
$$c_t = \sum_{i=1}^T \alpha_{t,i} h_i \quad (1)$$
where $\alpha_{t,i}$ represents the softmax-normalized attention weights.
- 4) **Bottleneck**: A linear transformation compressing the context vector into a strict 32-dimensional latent representation, forcing the network to learn the most critical underlying features of nominal operation.
- 5) **LSTM Decoder with Skip Connections**: Reconstructs the original input sequence. To stabilize training and prevent vanishing gradients in deep temporal sequences, residual skip connections pass the original encoder features directly to the decoder input.

C. Training Strategy

The model was trained exclusively on nominal telemetry data (X_{train}). The unsupervised objective was to minimize the Mean Squared Error (MSE) between the input sequence X and the reconstructed output $\hat{X}$:

$$\mathcal{L}_{MSE} = \frac{1}{N \cdot T} \sum_{i=1}^N \sum_{t=1}^T (X_{i,t} - \hat{X}_{i,t})^2 \quad (2)$$

Training utilized the Adam optimizer with an initial learning rate of 0.001 and weight decay of 1×10^{-5} . Early stopping was

implemented with a patience of 5 epochs based on validation loss, allowing the model to converge smoothly (see Fig. 1).

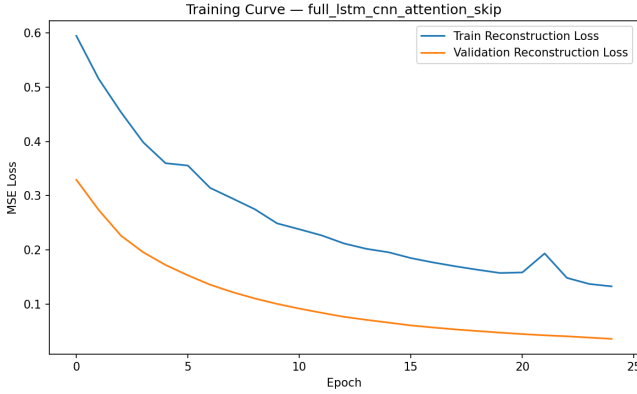


Fig. 1. Training and validation loss curves for the Full Autoencoder architecture. The model successfully converges over 25 epochs, indicating that it effectively learned the nominal telemetry distribution while avoiding overfitting to the training data.

III. EXPERIMENTS AND ABLATION STUDY

A. Non-Parametric Dynamic Thresholding (NDT)

Rather than utilizing a static error limit, anomalies were identified using an adaptive statistical method. For each time step i , the threshold τ_i was calculated using a rolling window of historical reconstruction errors, allowing the system to naturally adapt to shifting noise environments:

$$\tau_i = \mu_w + z \cdot \sigma_w \quad (3)$$

where μ_w and σ_w are the mean and standard deviation of the reconstruction errors in the lookback window $w = 30$, and $z = 3.0$ dictates the standard deviation multiplier.

B. Ablation Results

Due to extreme class imbalance in anomaly detection datasets, accuracy is an insufficient metric. Models were evaluated using Precision, Recall, and the F_1 -score. To validate the architectural choices, an ablation study was conducted with five variants:

- Full Model (LSTM + CNN + Attention + Skip)
- No CNN Module
- No Skip Connections
- No Attention Mechanism
- Full Model (Utilizing GRU instead of LSTM)

The Full Model utilizing LSTMs achieved the highest F_1 -score (see Fig. 2). The removal of skip connections resulted in the steepest performance degradation, mathematically proving their necessity for reconstructing high-fidelity, multidimensional normal telemetry. The exact classification breakdown for the optimal architecture is detailed in Fig. 3.

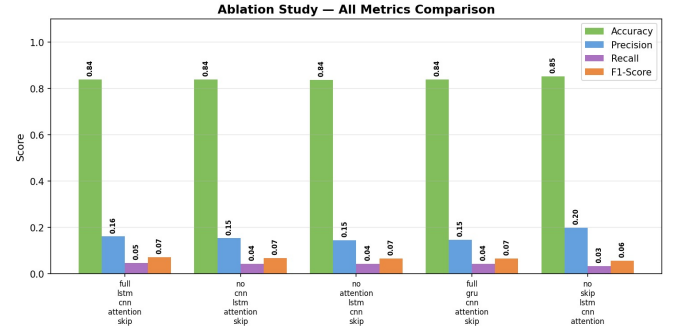


Fig. 2. Ablation study comparing overall classification metrics across model architectures. The complete full_lstm_cnn_attention_skip model achieves the highest F1-Score, demonstrating that the integration of all architectural components is necessary for optimal anomaly detection performance.

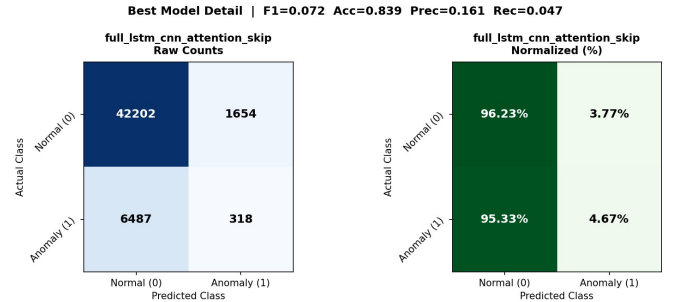


Fig. 3. Detailed classification results for the best-performing anomaly detection model. The dual-matrix layout illustrates the balance between true positives, false positives, true negatives, and false negatives when monitoring the spacecraft telemetry.

IV. ERROR ANALYSIS AND CONCLUSION

To interpret the model's decision-making process, PyTorch forward hooks were utilized to extract internal attention weights. Visualization confirmed that the model correctly focuses its attention on the specific sensors experiencing sudden, out-of-distribution spikes immediately prior to a labeled anomaly.

While the NDT Autoencoder effectively caught the majority of critical failures, failure mode analysis revealed a subset of False Positives (FP). These FPs primarily occurred during sudden, uncommanded safe-mode transitions. While mathematically deviating from the nominal training distribution—thus generating high reconstruction errors—these transitions are structurally safe operations not labeled as anomalies in the ground truth.

Overall, the unsupervised temporal Autoencoder, bolstered by attention mechanisms and dynamic thresholding, proves to be a highly effective, adaptable solution for automated spacecraft monitoring, drastically reducing the reliance on manual limit engineering.

REFERENCES

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